The Hierarchy for This Recap

1 Measurement	How does the world become data? Projection, sound waves, light transport
2 Data Structure	Which elements should interact? Sets, ordered sequences, harmonic neighbors
3 Size	Which information survives the budget? Spatial / temporal scales, tokens, latent codecs
4 Learning by prediction	What target teaches the desired structure? Contrastive pairs, masked targets, energies, scores
5 Composition and control	How can requirements constrain outputs? Relations, programs, conditioning, noise levels

PRINCIPLE 1

Measurement and Observation

Model the process that turns a physical scene into observations

3D Geometry and Projection

1 Measurement and observation

Structure: a pixel specifies a viewing ray; depth selects a 3D point

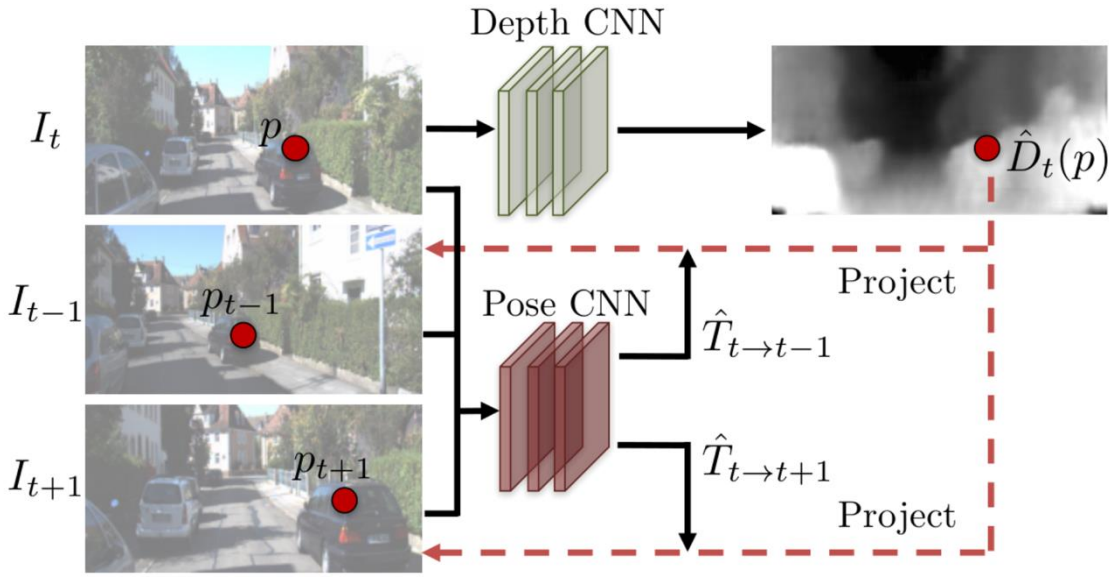
Rigid motion converts world coordinates into camera coordinates

Perspective projection divides by depth and removes one dimension

Depth and camera pose let us reproject one view into another

$$X_c = RX_w + t$$

$$u = \frac{f_x X_c}{Z_c} + c_x \quad v = \frac{f_y Y_c}{Z_c} + c_y$$



Depth and pose prediction with view reconstruction · L2 p24

Idea: use differentiable view reconstruction as supervision

Physical Structure in Signals

1 Measurement and observation

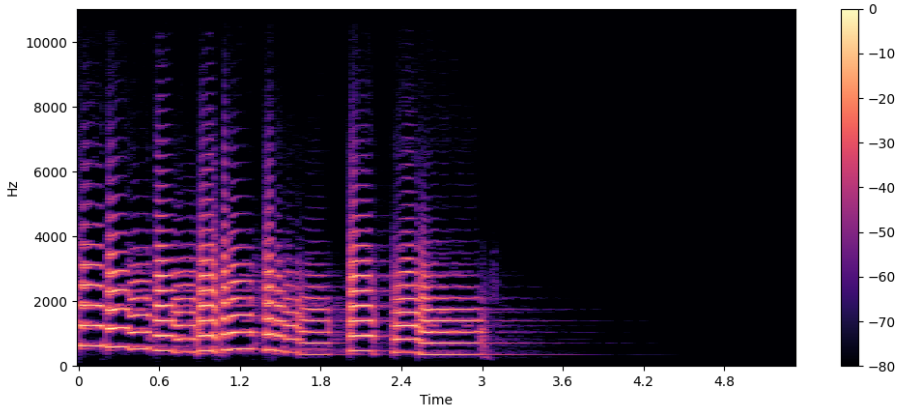
Meaning in practice: the forward process supplies useful algebra

Audio: a microphone records a time-varying pressure waveform

A short-time Fourier transform reveals local frequency content

Lighting: under fixed transport, contributions from lights add

$$I(L_1 + L_2) = I(L_1) + I(L_2)$$



Speech spectrogram · L8 p13

Ideas: spectral audio processing and light-transport consistency

Measurement: Key Ideas

1 Measurement and observation

Forward model	Specify projection, transformation, or signal formation
Inverse problem	Recover a hidden scene or source from partial observations
Learning idea	Reconstruct observations to obtain a consistency loss

Examples: depth + pose, inverse rendering, relighting

PRINCIPLE 2

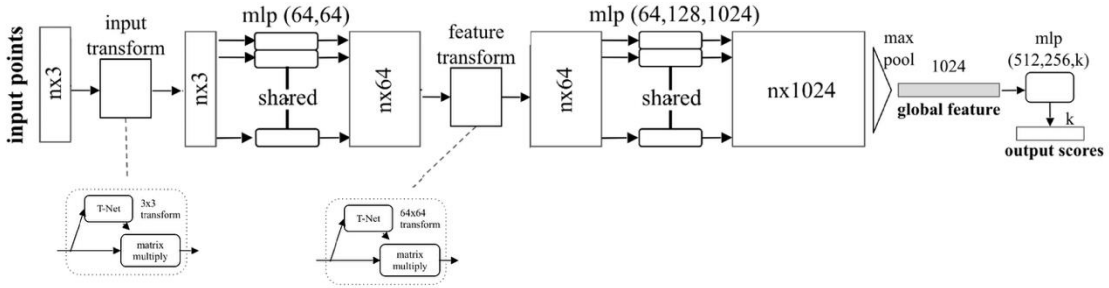
Data Structure

Make computation respect the invariance and equivariance

Sets and Sequences

2 Structure and interactions

3D points: order is arbitrary



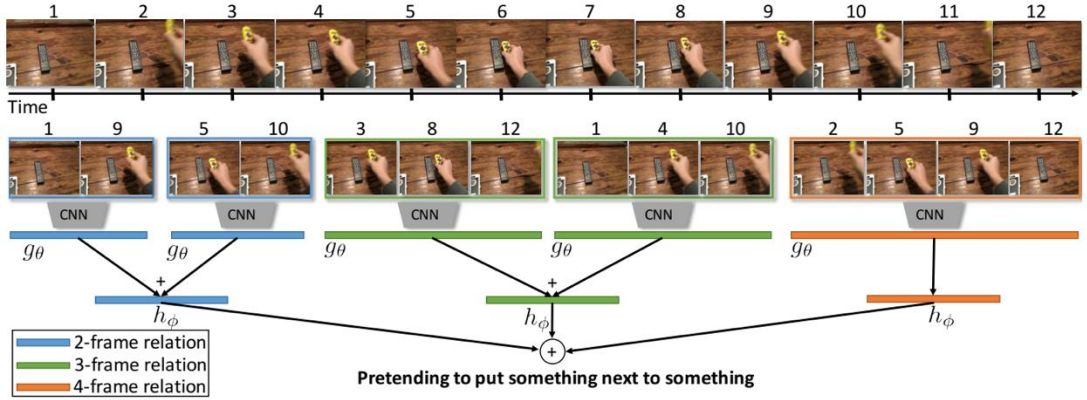
PointNet architecture · L3 p54

Reordering point storage leaves the same shape

Shared point functions plus symmetric pooling

Classification is invariant; point labels must reorder

Video frames: order carries meaning



Temporal Relation Networks · L10 p18

Opening and closing can contain similar frames

Frame averaging removes the order signal

Ordered temporal relations can distinguish events

The correct symmetry depends on the task and the output

Neighborhoods Follow the Modality

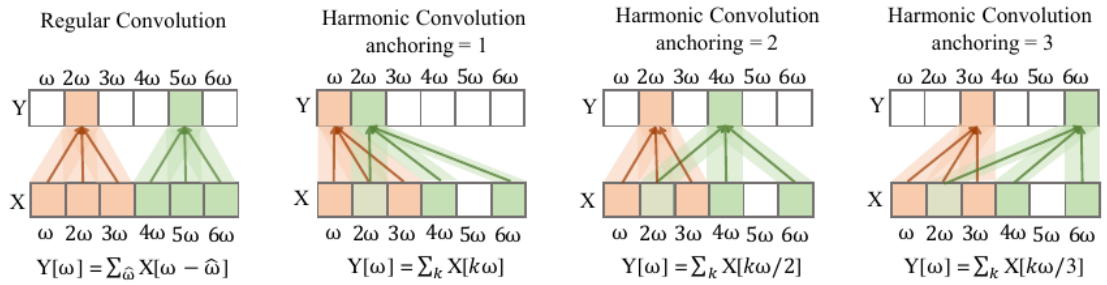
2 Structure and interactions

Structure: related elements need not be adjacent in the stored array

Image: nearby pixels support local pattern matching

Audio: harmonic multiples may belong to one source

Video: one moving object occupies different spatial positions



Regular and harmonic convolution · L8 p26

Ideas: convolution, harmonic kernels, flow, and non-local attention

Structure: Key Ideas

Preserve the right symmetry	Ignore point order; retain event order
Choose meaningful neighbors	Spatial distance, harmonic multiples,
Add context when needed	Use attention or global aggregation to resolve local ambiguity

Architecture specifies which relationships are easy to compute

PRINCIPLE 3

Size

Allocate information and computation at useful levels of detail

Multiresolution Across Modalities

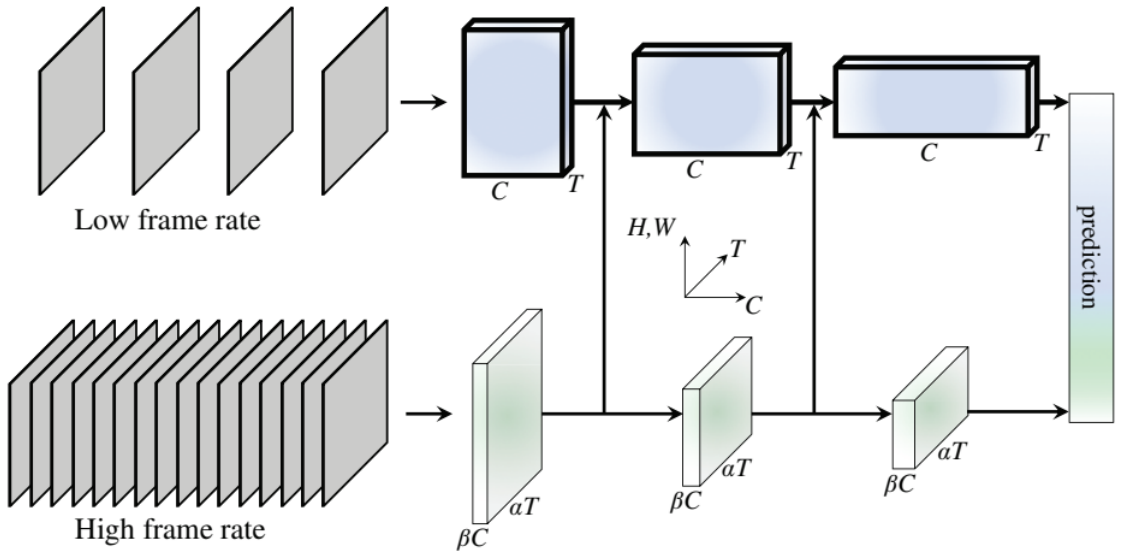
3 Scale and representation

Structure: useful detail occurs at different spatial and temporal scales

Images: small objects and precise boundaries need fine spatial detail

Point clouds: local groups build larger geometric neighborhoods

Video: SlowFast allocates different rates and widths to two pathways



SlowFast: low-rate appearance and high-rate motion pathways · L10 p29

Ideas: feature pyramids, PointNet++, and multirate computation

PRINCIPLE 4

Learning by Prediction

Contrastive Learning

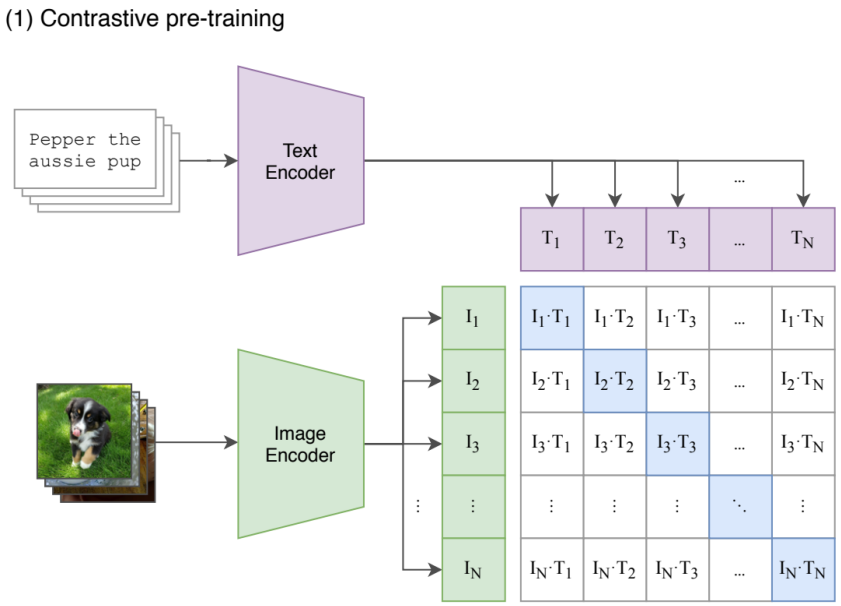
4 Learning by prediction

Meaning: identify the compatible observation among alternatives

Image–text: CLIP distinguishes paired examples within a batch

Speech: CPC identifies the real future latent among negatives

Compositional language: edited captions target object or relation errors



CLIP image–text contrastive pretraining · L5 p22

The negative examples determine which distinctions receive pressure

What Should Be Predicted?

Modality / method	Target	Structure encouraged
Language / VideoGPT	Next discrete token	Observed prefix and token ordering
HuBERT	Masked cluster label	Speech regularities beyond local copying
VideoMAE	Masked pixels	Visual reconstruction with tube masking
V-JEPA	Missing teacher features	Predictive structure in a learned space

Choose the target and the missing-information pattern together

Energy and Score Learning

4 Learning by prediction

Meaning: learn how good a candidate is, or how to improve it

$$p(x) = \frac{e^{-E(x)}}{Z} \qquad s(x) = \nabla_x \log p(x) = -\nabla_x E(x)$$

Detection: predict localization quality and optimize the box

Generation: an energy defines an unnormalized density

A score gives the local direction of increasing log density

Denoising learns this direction at multiple noise levels

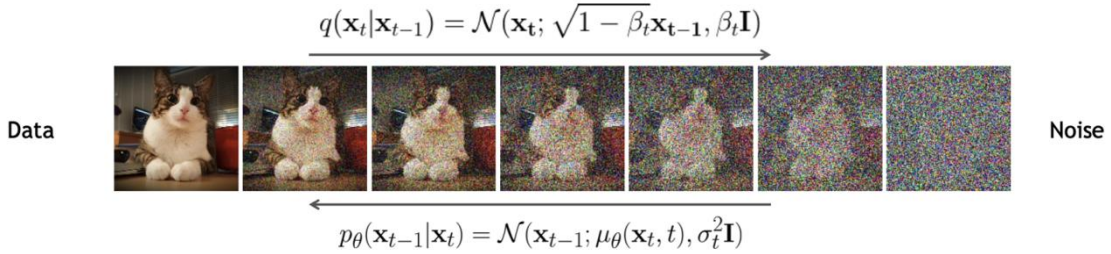


Image diffusion through a sequence of noise levels · L6 p36

Optimization seeks a solution; sampling must also represent uncertainty

Prediction: Key Ideas

Contrastive objectives	Reward a correct match relative to chosen alternatives
Completion objectives	Predict withheld tokens, pixels, labels, or teacher features
Energy / score objectives	Learn a candidate criterion or a denoising direction
Inference computation	Use iteration, sampling, or search when feedback is informative

Evaluate the behavior you wanted the prediction task to induce

PRINCIPLE 5

Composition and Control

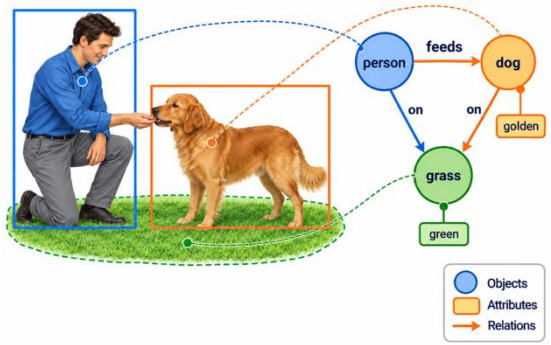
Meaning: express a complex requirement through parts and their dependencies

Objects, roles, and executable relationships
Conditioning paths inside a generator
Noise levels as information controls

Relations and Executable Structure

5 Composition and control

Language–vision relations



Scene-graph grounding · L5 p40

Objects and relations encode different facts

Grounding assigns each entity to a region

Matching all nouns can still reverse a relationship

3D programs



ShapeAssembly parameter edits · L7 p76

Attachments and symmetry link the parts

A parameter edit propagates through the program

The program vocabulary limits expressiveness

Explicit dependencies make the meaning of an edit easier to specify

Conditioning Needs a Computation Path

5 Composition and control

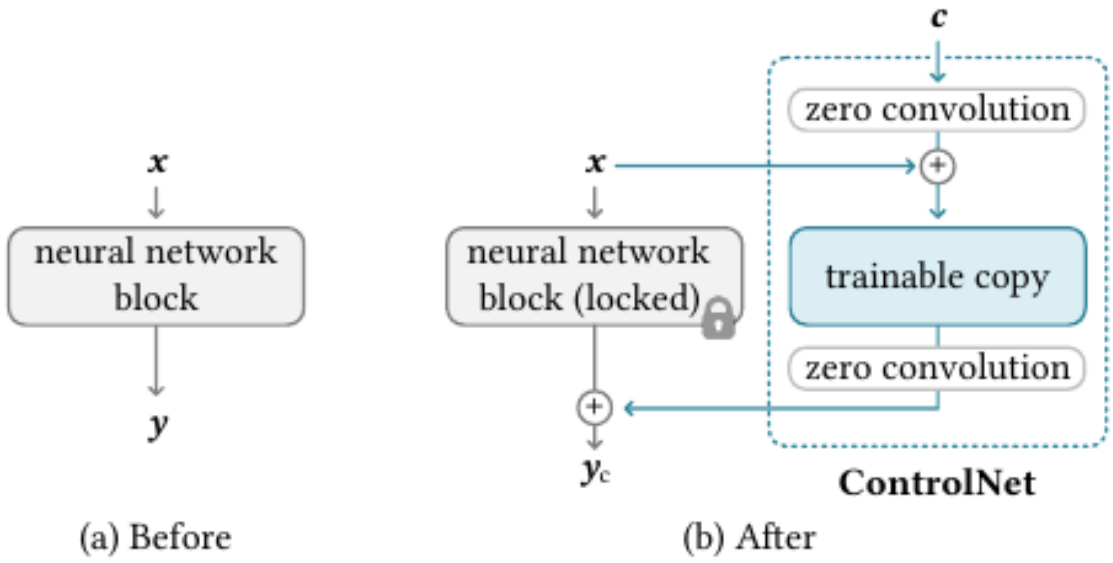
Structure: a requested change must reach the variables it should affect

Image layout: SPADE injects spatial scale and bias after normalization

ControlNet adds a learned residual path for spatial conditions

CameraCtrl encodes viewing rays; MotionCtrl separates motion controls

Music: AMT inserts specified events into the generated sequence



ControlNet: frozen generator with a trainable residual branch · L7 p52

Learned conditioning and explicit event insertion offer different guarantees

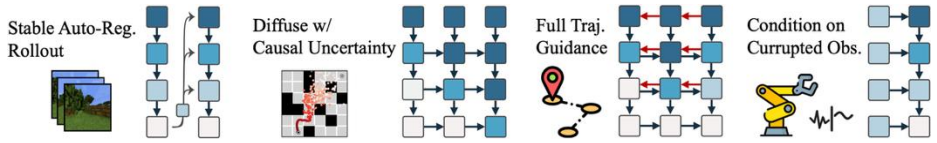
Noise as an Information Control

5 Composition and control

Meaning: noise level determines how much an observation constrains generation

$$x_t{}^k = \alpha_k x_t + \sigma_k \varepsilon_t$$

Clean	Observed content remains fully available	Condition on a known video frame
Partly noisy	Some evidence remains; larger changes become possible	Partial re-denoising for an image edit
Highly noisy	Very little original signal remains	Generate a missing or future frame



Diffusion Forcing inference schedules · L11 p47

Per-frame noise supports prediction, interpolation, and flexible histories

Highlight 1: Representations for Tasks

What does the task need to preserve, compare, reconstruct, or edit?

Task	Representation	Key characteristics
Recognition / retrieval	Global image, text, or clip features	Efficient similarity comparisons, compressed detail
3D geometry / interaction	Points, meshes, voxels, implicit fields	Different sampling, connectivity, memory, and query costs
Speech / music	Waveforms, spectra, codec codes, MIDI	Acoustic detail versus explicit note control
Language / grounding	BPE tokens, region features, scene graphs	Sequences, local correspondence, and directed relations
Generation / editing	VQ / VAE latents, factor codes, body skeletons, programs	Compression versus accessible editing variables

Representation choice depends on the task and the operations it needs

Highlight 2: Natural and Synthetic Data

Separate the origin of the observations from the construction of their targets

Source of supervision	Construction process	Examples and checks
Natural structure	Observe sequences, paired modalities, or shared views	Text continuation, image–text pairs, view reconstruction
Derived annotations	Record data, run a teacher, align and filter its labels	HuBERT clusters, video captions, poses, tracks
Generated editing pairs	Edit captions, couple image generation, filter candidate pairs	InstructPix2Pix: check the change and preservation
Rendering / programs	Sample parameters, execute a change, render paired outputs	IC-Light relighting and Pattern Analogies quartets

Trace the pair construction, its assumptions, and its remaining errors

Highlight 3: Inference as Computation

Closing highlights

Specify what changes at inference and which feedback guides each update

Technique	What the computation does	Representative mechanisms
Conditioning / guidance	Route evidence into the model and tune its influence	Classifier-free guidance
Compositional inference	Combine compatible factors on the same candidate	Energy / score composition
Iterative correction	Critique a result and partially re-denoise it	Revised prompts
Noise / history schedules	Choose which frames provide evidence and which are generated	Diffusion Forcing and history-guided video diffusion

More denoising steps, MCMC, and critique loops perform different computations